

Rupin Khandelwal

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Professional Summary

Results-driven Technical Leader & Solution Architect with 15+ years of international experience designing and delivering highly scalable, secure, and low-latency systems across banking, payments, trading, and consumer platforms. Expert in cloud-native architectures, microservices, and mobile-first platforms, with a strong track record of owning complex system design, solving hard engineering problems, and leading technical execution end to end. Known for combining deep technical expertise with pragmatic leadership to deliver systems used by millions of users across multiple countries.

- **End-to-End System Design:** Ownership of architecture, design decisions, and implementation across mobile, backend, and cloud platforms.
- **Cloud & Distributed Systems:** Designed and scaled AWS-based microservices handling high transaction volumes with strong security and reliability.
- **Mobile-at-Scale Architecture:** Built mobile platforms with BFF layers, caching, and API orchestration supporting millions of active users.
- **Security & Compliance:** Designed and implemented security-sensitive systems including MFA, encryption, secure fund transfers, and regulatory-compliant workflows
- **Performance Engineering:** Deep focus on latency, throughput, and reliability; consistently improved performance and reduced response times
- **Technical Mentorship:** Acted as technical authority for teams (8–10 engineers), guiding design, reviewing critical code paths, and enforcing engineering standards

Core Skills

- **Architecture & Cloud:** AWS (Certified), Microservices, Distributed Systems, Event-Driven Systems, BFF Pattern, Caching, Security Architecture
- **Backend Engineering:** Java 17, Spring Boot, REST APIs, JPA, High-Performance Services
- **Mobile Engineering:** Native iOS & Android, Mobile Architecture, Firebase, Secure Mobile SDKs
- **Programming Languages:** Java, Swift, C/C++, Objective-C, SQL
- **DevOps & Delivery:** CI/CD, Release Automation, Production Monitoring
- **Engineering Practices:** HLD/LLD, Code Reviews, Performance Tuning, Technical Design Reviews
- **Domains:** Retail & Corporate Banking, Digital Payments, Capital Markets, Algo Trading

Certifications

- AWS Certified Solutions Architect – Associate (Amazon Web Services, 2022)

Professional Experience

HCL Technologies, Singapore | Technical Architect

Mar 2021 – Jan 2026

Project: UOB Infinity – Corporate banking

- Owned architecture and technical execution for large-scale mobile and backend systems within UOB's retail and corporate banking platforms
- Collaborated with product managers, designers, and stakeholders to define technical requirements and project scope.
- Acted as technical decision-maker, driving design reviews, resolving complex production issues, and guiding teams on scalability and security.
- Conducted code reviews, ensuring adherence to coding standards and best practices.
- Collaborated with cross-functional teams to resolve technical challenges and optimize system performance.

Project: UOB TMRW – Retail Banking

- Led & Develop full-stack feature delivery across Mobile Apps (iOS & Android) and Backend (Java, Spring Boot & Microservices) for one of Singapore's leading retail banking apps, used by millions of customers.
- Managed and executed release development for cross-country deployments, including Malaysia and Thailand.
- Implemented CI/CD pipelines using Fastlane & GitHub Actions.
- Spearheaded fund transfers, 3DS card authentication, card services integration, transaction security modules like OFM, MFA and multi-level encryption, reducing fraud risk and enhancing compliance.
- Conducted performance optimization, reducing app load times by 30%, boosting customer satisfaction.
- Achieved 20% faster project delivery, helping UOB launch features ahead of competitors.

Nityo InfoTech, Singapore | Senior Technical Lead

Nov 2020 – Mar 2021

Project: DBS DigiBank App – Retail Banking App (Singapore)

- Delivered modernization of DBS's flagship digital banking app, ensuring compliance with regulatory & SLA requirements.
- Introduced reusable components & cloud-hosted APIs, improving scalability and cutting dev time for future releases.
- Collaborated with business & compliance teams, ensuring technical solutions met regulatory mandates.

Optimum Solutions, Singapore | Technical Lead

Mar 2019 – Nov 2020

Project: UOB Bizz App

- Designed and delivered enterprise-grade business banking app with continuous delivery pipelines.
- Analyzed client requirements, created estimates, and managed agile sprints to ensure timely delivery.
- Improved live app performance by 25% through code optimizations and quality upgrades.
- Acted as technical owner, ensuring end-to-end delivery across multiple teams.

Capgemini, Sweden & India | Technical Lead / Senior Developer

Jun 2014 – Feb 2019

Project: Husqvarna Fleet Services 2.0 (Sweden)

- Developed mobile app from scratch for fleet & equipment management.
- Owned system design, implementation, and app quality upgrades, enabling efficient field operations.
- Delivered project ahead of schedule, contributing to faster rollout in Europe.

Project: Coca-Cola Freestyle Machine (USA)

- Built UI interfaces and mobile integrations for Coca-Cola's smart dispenser machines.
- Developed in C/C++, Qt, Swift and QNX ensuring smooth operations in thousands of locations worldwide.
- Contributed to automation tools (ATA, CDM, cartridge management), reducing maintenance costs.

Tata Consultancy Services (TCS), India | System Engineer

Mar 2012 – Jun 2014

Projects: Travel Assist & Customer Insights

- Developed iOS apps in Objective-C for corporate travel & customer analytics.
- Led development + test teams, ensuring quality delivery under agile frameworks.
- Partnered with UX/UI teams to create user-friendly interfaces.

Dataman Computer Systems, India | Software Developer

Aug 2009 – Sep 2011

Project: Smart Edge Algo Trading Application

- Built algo trading app with hedging strategies, handling live trades & maintaining positions in real time.
- Developed in C/C++, VC++ & SQL Server, delivering reliable performance in stock trading environments.

Project: Automated Arbitrage Algo Trading System

- Created exchange interfaces for BSE, NSE, MCX, DGCX, handling large-scale communication protocols (FIX, Structural, Fast FIX).
- Deployed system at Kotak Securities (Mumbai), improving trading efficiency for institutional clients.

Education

- Master of Computer Application (MCA) – Uttar Pradesh Technical University, India (2006–2009)
- Bachelor of Science in Electronics – C.S.J.M University, India (2002–2005)

Portfolio (Live Projects)

- UOB TMRW: <https://apps.apple.com/sg/app/uob-tmrw/id1049286296>
- DBS DigiBank: <https://apps.apple.com/sg/app/dbs-digibank/id1068403826>
- Husqvarna Fleet Services 2.0: <https://apps.apple.com/us/app/husqvarna-fleet-services-2-0/id1334672726>